PAYAL KUMARI

Bengaluru, India | itsmepayal123@gmail.com | github.com/payal-315
linkedin.com/in/payal315 | https://payal-portfolio-1.vercel.app/

TECHNICAL SUMMARY

Aspiring Software Engineer with a strong foundation in Computer Science (Cyber Security) and a **9.19 CGPA**. Proficient in developing **secure, high-quality code** in Java and Python. Experienced in **CI/CD**, cloud technologies, and relational databases, with familiarity in **Agile** and SDLC.

EDUCATION

Bachelor of Engineering in CSE (Cyber Security) — Expected May 2027
Sri Venkateshwara College Of Engineering, Bangalore — CGPA: 9.19
Relevant Coursework: Data Structures & Algorithms, Discrete Mathematics, OOP, DBMS, OS.

SKILLS

Languages	Java (Advanced OOP), Python, SQL (MySQL), C, Shell Scripting
Frameworks/Tools	React, Node.js, HTML5, CSS3, TensorFlow, Keras, PyTest, JUnit, Docker, Linux CLI
Cloud & DevOps	Microsoft Azure, GitHub Actions, CI/CD Automation, Agile
Technical Core	DSA, SDLC, Secure Coding, System Reliability, Observability
Data Engineering	Relational Database Design, Indexing, Big Data Concepts

EXPERIENCE

Software Developer Intern | Crime Investigation Department (CID) — Aug 2025 – Sept 2025
Karnataka, India

- **Collaborative Software Development:** Worked in a team to design and develop a **Deepfake Detection system for multimedia** in digital forensics, contributing to secure and high-quality code for real-world investigative use.
- **Technical Troubleshooting & Data Analysis:** Analyzed and synthesized forensic datasets to identify performance bottlenecks; applied **performance profiling** and asynchronous processing to improve execution efficiency by **25%**.
- **System Reliability & Application Resiliency:** Reduced manual toil by implementing structured logging and debugging workflows, improving system stability, monitoring, and overall **SDLC efficiency**.

PROJECTS

- **Deepfake Detection System** | ***Python, TensorFlow, PyTest*** — *[Link]*
  Engineered a CNN-based system for real-time analysis; used **automated testing** to ensure 95% inference reliability across diverse datasets and maintained **high-quality code** standards.
- **EduMatrix-Portal: Relational Data Systems** | ***SQL, MySQL*** — *[Link]*
  Architected a relational schema with complex constraints to ensure **data integrity** for academic tracking; optimized queries via indexing to reduce retrieval latency by **40%** and synthesized data sets for **technical troubleshooting**.
- **Modular Game Engine: Tic-Tac-Toe Ultra** | ***Java (Advanced OOP)*** — *[Link]*
  Designed a modular logic engine using **Polymorphism and Encapsulation** to separate core game mechanics from UI; applied **Test-Driven Development (TDD)** to validate logic and memory efficiency, ensuring a stable and **resilient software architecture**.

ACHIEVEMENTS & LEADERSHIP

- **3rd Prize Winner, CIDECODE Hackathon:** Awarded by **CID Karnataka** for developing an AI-driven Deepfake Detection solution; recognized for excellence in problem-solving and secure software engineering.
- **Smart India Hackathon (SIH) Participant:** Developed a "Smart Irrigation System" using a logical system architecture to provide sustainable technology solutions for social impact.
- **Admin, Checkmate Chess Club:** Directed operations for 100+ members and coordinated large-scale events, demonstrating **leadership and organizational skills** in a fast-paced environment. *[Website]*